

Spencer Teetaert, PhD Student

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🌐 <https://spencerteetaert.github.io/>

Research Interests

My research focuses on stochastic dynamic state estimation for continuum and soft robotic systems, with an emphasis on continuous-time probabilistic methods for highly compliant structures operating under uncertainty. I aim to enable robust deployment of continuum robots in real-world, unstructured environments beyond controlled laboratory settings.

Education

- 2023 – present 📖 **Ph.D. Student in Robotics Engineering, University of Toronto** under the supervision of Dr. Timothy D. Barfoot and Dr. Jessica Burgner-Kahrs. Research focus on stochastic state estimation for continuum robots.
- 2018 – 2023 📖 **B.A.Sc. Engineering Science, University of Toronto** in Robotics. GPA: 3.77 / 4.0
Thesis: *Towards Learning-Based Control for a Tendon-Driven Parallel Continuum Robot.*

Honours and Awards

- 2025-26 📖 **University of Toronto Robotics Institute Fellow**
 📖 **Queen Elizabeth II Graduate Scholarship in Science and Technology**
- 2024-25 📖 **Ontario Graduate Scholarship**
- 2022 📖 **NSERC Undergraduate Student Research Award**
- 2018 📖 **Faculty of Applied Science & Engineering Admission Scholarship**
 📖 **Dr. Dale Iwanoczko Memorial Scholarship**

Publications

Peer-Reviewed Journal Articles

- 1 **S. Teetaert**, W. Zhao, A. Loquercio, S. Zhou, L. Brunke, M. Schuck, W. Hönig, J. Panerati, and A. P. Schoellig, “Advancing reproducibility, benchmarks, and education with remote sim2real: Remote simulation to real robot hardware,” *IEEE Robotics & Automation Magazine*, vol. 32, no. 1, pp. 117–123, 2025. 🌐 DOI: 10.1109/MRA.2025.3527291.

Conference Papers

- 1 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “A sliding-window filter for online continuous-time continuum robot state estimation,” in *2026 IEEE 9th International Conference on Soft Robotics (RoboSoft)*, 2026, forthcoming.
- 2 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, “Space-time continuum: Continuous shape and time state estimation for flexible robots,” 40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA40), 2024.

Preprints

- 1 **S. Teetaert**, G. Caroleo, M. Pontin, S. Lilge, J. Burgner-Kahrs, T. D. Barfoot, and P. Maiolino, *Continuum robot localization using distributed time-of-flight sensors*, Submitted to *Robotics Science and Systems (RSS)*, 2026. arXiv: 2602.07209.

- 2 **S. Teetaert***, E. Walsh*, E. Diller, T. D. Barfoot, and J. Burgner-Kahrs, *Sliding sensors: Configurable confidence in state estimation for continuum robots*, Submitted to 2026 IEEE 9th International Conference on Soft Robotics (RoboSoft), 2026.
- 3 **S. Teetaert**, S. Lilge, J. Burgner-Kahrs, and T. D. Barfoot, *A stochastic framework for continuous-time state estimation of continuum robots*, In revision at IEEE Transactions on Robotics., 2025. arXiv: 2510.01381.
- 4 **S. Teetaert**, W. Zhao, N. Xinyuan, H. Zahir, H. Leong, M. Hidalgo, G. Puga, T. Lorente, N. Espinosa, J. A. D. Carrasco, K. Zhang, J. Di, T. Jin, X. Li, Y. Zhou, X. Liang, C. Zhang, A. Loquercio, S. Zhou, L. Brunke, M. Greeff, W. Hoenig, J. Panerati, and A. P. Schoellig, *A remote sim2real aerial competition: Fostering reproducibility and solutions' diversity in robotics challenges*, 2023. arXiv: 2308.16743.

Employment History

Teaching Experience

- 2024–present  **ROB498:** Practical teaching assistant for engineering undergraduate Robotics Capstone Design. Head TA as of 2026. Assists in course design and technical aid to students building custom drones.
- 2023  **ROB301:** Marking teaching assistant for undergraduate Introduction to Robotics.

Industry Experience

- 2021–2022  **Research Engineer Coop**, Human Machine Interaction Lab, Huawei Canada. Developed advanced machine learning computer vision models for eye-gaze and gesture tracking with applications to consumer electronic devices.
- 2019–2020  **Industrial Engineering Coop**, Hylife Foods
Launch an automation overhaul project for a production plant of 2,000+ workers. Built relationships with third party robotics partners and aided in design projects. Lead a cutting blade optimization project.

Service

Professional

- 2022  **Co-organizer**, Safe Robot Learning Competition, 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Extra-Curriculars and Volunteer

- 2025–present  **University of Toronto Undergraduate Research Students' Association**
• Graduate Student Mentor
- 2020–2023  **University of Toronto Robotics Association**
• President
• Project Manager, Autonomous Rover Team
• Computer Vision Designer, Autonomous Rover Team
- 2019–2023  **University of Toronto Engineering Orientation**
• Subcommittee Chair, Photography
• Subcommittee Chair, Retreat
• Group Leader